数学准备

$\nabla\cdot(\varphi\vec f)=(\nabla\varphi)\cdot\vec f+\varphi\nabla\cdot\vec f$ $\nabla\times(\varphi\vec f)=(\nabla\varphi)\times\vec f+\varphi\nabla\times\vec f$

$\nabla\cdot(\vec f\times\vec g)=(\nabla\times\vec f)\cdot\vec g-\vec f\cdot(\nabla\times\vec g)$

$\nabla\times(\vec f\times\vec g)=(\vec g\cdot\nabla)\vec f+(\nabla\cdot\vec g)\vec f-(\vec f\cdot\nabla)\vec g-(\nabla\cdot\vec f)\vec g$

$\nabla(\vec f\cdot\vec g)=\vec f\times(\nabla\times\vec g)+(\vec f\cdot\nabla)\vec g+\vec g\times(\nabla\times\vec f)+(\vec g\cdot\nabla)\vec f$

$\nabla\times(\nabla\times\vec f)=\nabla(\nabla\cdot\vec f)-\nabla^2\vec f$

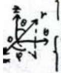
柱坐标系 $\begin{cases}\hat e_\rho=\hat e_x\cos\phi+\hat e_y\sin\phi\\ \hat e_\phi=-\hat e_x\sin\phi+\hat e_y\cos\phi\end{cases}$ $\begin{cases}\hat e_x=\hat e_\rho\cos\phi-\hat e_\phi\sin\phi\\ \hat e_y=\hat e_\rho\sin\phi+\hat e_\phi\cos\phi\end{cases}$

$\nabla\psi=\hat e_\rho\frac{\partial\psi}{\partial\rho}+\hat e_\phi\frac{1}{\rho}\frac{\partial\psi}{\partial\phi}+\hat e_z\frac{\partial\psi}{\partial z}$

$\nabla^2\psi=\frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho\frac{\partial\psi}{\partial\rho})+\frac{1}{\rho^2}\frac{\partial^2\psi}{\partial\phi^2}+\frac{\partial^2\psi}{\partial z^2}$ $\nabla\times\vec A=\frac{1}{\rho}\begin{vmatrix}\hat e_\rho & \rho\hat e_\phi & \hat e_z\\ \frac{\partial}{\partial\rho} & \frac{\partial}{\partial\phi} & \frac{\partial}{\partial z}\\ A_\rho & \rho A_\phi & A_z\end{vmatrix}$

$\nabla\cdot\vec A=\frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho A_\rho)+\frac{1}{\rho}\frac{\partial A_\phi}{\partial\phi}+\frac{\partial A_z}{\partial z}$

球坐标系 $\begin{cases}\hat e_r=\hat e_x\sin\theta\cos\varphi+\hat e_y\sin\theta\sin\varphi+\hat e_z\cos\theta\\ \hat e_\theta=\hat e_x\cos\theta\cos\varphi+\hat e_y\cos\theta\sin\varphi-\hat e_z\sin\theta\\ \hat e_\varphi=-\hat e_x\sin\varphi+\hat e_y\cos\varphi\end{cases}$



$\begin{cases}\hat e_x=\sin\theta\cos\phi\,\hat e_r+\cos\theta\cos\phi\,\hat e_\theta-\sin\phi\,\hat e_\phi\\ \hat e_y=\sin\theta\sin\phi\,\hat e_r+\cos\theta\sin\phi\,\hat e_\theta+\cos\phi\,\hat e_\phi\\ \hat e_z=\cos\theta\,\hat e_r-\sin\theta\,\hat e_\theta\end{cases}$ $\nabla\cdot(\frac{\vec r}{r^3})=4\pi\delta(\vec x-\vec x')$ $\nabla\times(\frac{\vec r}{r^3})=0$

$\nabla\psi=\frac{\partial\psi}{\partial r}\hat e_r+\frac{1}{r}\frac{\partial\psi}{\partial\theta}\hat e_\theta+\frac{1}{r\sin\theta}\frac{\partial\psi}{\partial\phi}\hat e_\phi$

$\nabla^2\psi=\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial\psi}{\partial r})+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial\psi}{\partial\theta})+\frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}$ $\nabla\frac{1}{R}=-\frac{\vec R}{R^3}$

$\nabla\cdot\vec A=\frac{1}{r^2}\frac{\partial}{\partial r}(r^2 f_r)+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta f_\theta)+\frac{1}{r\sin\theta}\frac{\partial f_\phi}{\partial\phi}$

$\nabla\times\vec A=\frac{1}{r^2\sin\theta}\begin{vmatrix}\hat e_r & r\hat e_\theta & r\sin\theta\hat e_\phi\\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi}\\ A_r & rA_\theta & r\sin\theta A_\phi\end{vmatrix}$ $\rho_P=-(1-\frac{\varepsilon_0}{\varepsilon})\rho_f$ $\vec J_M=(\frac{\mu}{\mu_0}-1)\vec J_f$

电磁现象基本规律

$\vec E(\vec x)=\int_V\frac{\rho(\vec x')\vec r\,dV'}{4\pi\varepsilon_0 r^3}$ $\nabla\cdot\vec E(\vec x)=\frac{\rho(\vec x)}{\varepsilon_0}$, $\nabla\times\vec E=0$ $\oint\vec E\cdot d\vec S=\frac{Q}{\varepsilon_0}$

$\vec E=-\nabla\varphi$. $\nabla^2\varphi=-\frac{\rho}{\varepsilon_0}$

$\vec J=\rho\vec v$ $I=\int\vec J\cdot d\vec S$ $\nabla\cdot\vec J+\frac{\partial\rho}{\partial t}=0$ $\oint\vec J\cdot d\vec S=-\int\frac{\partial\rho}{\partial t}dV$.

$\vec B=\frac{\mu_0}{4\pi}\int_V\frac{\vec J(\vec x')\times\vec r}{r^3}dV'$ $\nabla\cdot\vec B=0$, $\nabla\times\vec B=\mu_0\vec J$. $\oint_L\vec B\cdot d\vec l=\mu_0 I$

$\nabla\cdot\vec E=\frac{\rho}{\varepsilon_0}$. $\nabla\times\vec E=-\frac{\partial\vec B}{\partial t}$, $\nabla\cdot\vec B=0$. $\nabla\times\vec B=\mu_0\vec J+\mu_0\varepsilon_0\frac{\partial\vec E}{\partial t}$.

$\vec J=\sigma\vec E$ 单位体积电场功耗 $\frac{dW}{dt}=\vec J\cdot\vec E=\sigma E^2$

$\vec P=\chi_e\varepsilon_0\vec E$. $\vec M=\chi_m\vec H$, $\frac{\partial\vec P}{\partial t}=\vec J_p$, $\nabla\cdot\vec J_p+\frac{\partial\rho_p}{\partial t}=0$.

$\oint\vec D\cdot d\vec S=\int\rho_f dV=Q$, $\oint_L\vec E\cdot d\vec l=-\frac{d}{dt}\iint\vec B\cdot d\vec S$,

$\oint\vec B\cdot d\vec S=0$. $\oint\vec H\cdot d\vec l=\int\vec J_f\cdot d\vec S+\frac{d}{dt}\int\vec D\cdot d\vec S=I_f+\frac{d}{dt}\iint\vec D\cdot d\vec S$

$\hat e_n\cdot(\vec D_2-\vec D_1)=\sigma_f$, $\nabla\cdot\vec D=\rho_f$ $-\hat e_n\cdot(\vec P_2-\vec P_1)=\sigma_P$, $\nabla\cdot\vec P=-\rho_P$.

$\hat e_n\cdot(\vec E_2-\vec E_1)=\frac{\sigma_P+\sigma_f}{\varepsilon_0}$, $\nabla\cdot\vec E=\frac{\rho_f+\rho_P}{\varepsilon_0}$ $\hat e_n\cdot(\vec B_2-\vec B_1)=0$.

$\hat e_n\times(\vec H_2-\vec H_1)=\vec\alpha_f$, $\hat e_n\times(\vec E_2-\vec E_1)=0$, $\hat e_n\times(\vec M_2-\vec M_1)=\vec\alpha_M$, $\vec J_M=\nabla\times\vec M$

理想导体 $\sigma=\infty$: $\hat e_n\times\vec H=\vec\alpha$, 若 $\vec D=\varepsilon_0$... 单位时间通过界面流入 V内电磁场能量等于V内电...

理想介质 $\sigma=0$: $\vec\alpha=0$, $\sigma_f=0$ 对电荷做功功率与V内电磁场能量...

$-\oint\vec S\cdot d\vec S=\int\vec f\cdot\vec v\,dV+\frac{d}{dt}\int w\,dV$ $\nabla\cdot\vec S+\frac{\partial w}{\partial t}=-\vec f\cdot\vec v=\vec J\cdot\vec E$ 电磁力密度

$\vec S=\vec E\times\vec H$. $w=\frac12(\varepsilon_0E^2+\frac{1}{\mu_0}B^2)$ $\to\vec f=\vec E\rho+\vec J\times\vec B$

$\frac{\partial\rho}{\partial t}=-\nabla\cdot\vec J=-\frac{\sigma}{\varepsilon}\nabla\cdot\vec D=-\frac{\sigma\rho}{\varepsilon}$. $\rho=\rho_0e^{-\frac{\sigma}{\varepsilon}t}$ 衰减特征时间 $\tau=\frac{\varepsilon}{\sigma}$

静电场

$\vec E=-\nabla\varphi$ $\varphi(P_2)-\varphi(P_1)=-\int_{P_1}^{P_2}\vec E\cdot d\vec l$ $\varphi(P)=\int_P^\infty\vec E\cdot d\vec l$

点电荷 $\varphi=\frac{Q}{4\pi\varepsilon r}$ 分布电荷 $\varphi=\frac{1}{4\pi\varepsilon_0}\int\frac{\rho(\vec x')}{r}dV'+C$

$\nabla^2\varphi=-\frac{\rho_f}{\varepsilon}$ 介质分界面 $\varphi_1=\varphi_2$, $\varepsilon_2\frac{\partial\varphi_2}{\partial n}-\varepsilon_1\frac{\partial\varphi_1}{\partial n}=-\sigma_f$.

n: 1指向2. 导体表面 $\varphi_S=\text{const}$. $\varepsilon\frac{\partial\varphi}{\partial n}|_S=-\sigma_f$ (导体)

$\varepsilon_0(\frac{\partial\varphi_2}{\partial n}-\frac{\partial\varphi_1}{\partial n})=-(\sigma_f+\sigma_P)$ $\oint_S\varepsilon\frac{\partial\varphi}{\partial n}dS=-Q_0$ n: 导→外

均匀电场 $\varphi(P)=\varphi_0-\vec E_0\cdot\vec x$

电偶极矩 $\varphi=\frac{\vec p\cdot\vec r}{4\pi\varepsilon_0r^3}=\frac{ql\cos\theta}{4\pi\varepsilon_0r^2}$. $\vec E=\frac{1}{4\pi\varepsilon_0}(\frac{3(\vec p\cdot\vec r)\vec r}{r^5}-\frac{\vec p}{r^3})$

带电无限长导线 $\varphi(P)-\varphi(P_0)=\frac{\tau}{2\pi\varepsilon_0}\ln\frac{R_0}{R}$, $E_R=\frac{\tau}{2\pi\varepsilon_0R}$, $E_\theta=E_z=0$.

①直角坐标系 $\varphi(x,y,z)=X(x)Y(y)Z(z)$

$\begin{cases}\frac{d^2X(x)}{dx^2}+\alpha X(x)=0\\ \frac{d^2Y(y)}{dy^2}+\beta Y(y)=0\\ \frac{d^2Z(z)}{dz^2}+\gamma Z(z)=0\end{cases}$ $\alpha+\beta+\gamma=0$ 若与z无关 $\varphi=X(x)Y(y)$ $\begin{cases}\frac{d^2X}{dx^2}+\alpha X=0\\ \frac{d^2Y}{dy^2}+\beta Y=0\end{cases}$ $\alpha+\beta=0$ 若与y,z均无关 $\varphi=\varphi(x)$ $\varphi=Ax+B$ 均匀电场

②柱坐标系 $\varphi(r,\theta,z)=R(r)\Theta(\theta)$ 通常与z无关.

$\varphi(r,\theta)=A_0+\sum_{n=1}^\infty(A_nr^n+B_nr^{-n})(C_n\cos n\theta+D_n\sin n\theta)$

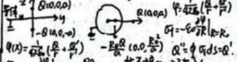
B_0 $\ln r$ 项去掉 $\varphi|_{r=0}$ 有限 (无中心线电荷) 可以去掉.

③球坐标系 $\varphi(r,\theta,\phi)=R(r)Y(\theta,\phi)$

轴对称 $\varphi(r,\theta)=\sum_{n=0}^\infty(A_nr^n+\frac{B_n}{r^{n+1}})P_n(\cos\theta)$.

球对称 $\varphi(r)=A+\frac{B}{r}$ 未知电荷进行求解.

镜像法: 用一个或几个假想点电荷等效代替实际的(感应电荷)



$\varphi(x)=\frac{1}{4\pi\varepsilon_0}(\frac{Q}{r}+\frac{Q'}{r'})$ $-\frac{R_0}{a}Q$ $(0,0,\frac{R_0^2}{a})$ $Q''=\oint\sigma_i dS=Q'$.

$\sigma_i=-\varepsilon_0\frac{\partial\varphi}{\partial z}|_{z=0}=-\frac{Qa}{2\pi r_1^3}$. 若不接地, 还有 Q'.

$Q''=\int\sigma_i dS=-Q=Q'$. (导体) $\varphi=\frac{1}{4\pi\varepsilon_0}(\frac{Q}{r}+\frac{Q'}{r'})+\frac{R_0Q+Q'}{4\pi\varepsilon_0R}$.

导体 $\varepsilon\to\infty$.

$\varphi(x)=\frac{1}{4\pi\varepsilon_0}[\frac{Q}{R}-\vec p\cdot\nabla\frac1R+\frac16\sum_{ij}D_{ij}\frac{\partial^2}{\partial x_i\partial x_j}\frac1R+\cdots]$

$\varphi^{(0)}=\frac{Q}{4\pi\varepsilon_0R}$ 所有电荷集中在原点激发电势.

$\varphi^{(1)}=-\frac{\vec p\cdot\nabla\frac1R}{4\pi\varepsilon_0}=\frac{\vec p\cdot\vec R}{4\pi\varepsilon_0R^3}$ $\vec p=\int_V\rho(\vec x')\vec x'dV'$ (原点在 ε 对称时为0)

$\varphi^{(2)}=\frac{1}{24\pi\varepsilon_0}\sum_{ij}D_{ij}\frac{\partial^2}{\partial x_i\partial x_j}\frac1R$. $D_{ij}=\int(3x_i'x_j'-r'^2\delta_{ij})\rho(\vec x')dV'$.

$D_{11}+D_{22}+D_{33}=0$ 仅5个独立分量. 球对称无电四极矩.

电磁波传播

$\nabla^2\vec E-\varepsilon_0\mu_0\frac{\partial^2\vec E}{\partial t^2}=0$ $\nabla^2\vec B-\varepsilon_0\mu_0\frac{\partial^2\vec B}{\partial t^2}=0$, $\begin{cases}\nabla\cdot\vec E=i\vec k\cdot\vec E\\ \nabla\times\vec E=i\vec k\times\vec E\end{cases}$ 平面时谐

时谐波: $\nabla\cdot\vec E=0$, $\nabla\times\vec E=i\omega\mu\vec H$. $\nabla\cdot\vec H=0$. $\nabla\times\vec H=-i\omega\varepsilon\vec E$.

Helmholtz: $\nabla^2\vec E+k^2\vec E=0$. $\nabla^2\vec B+k^2\vec B=0$ $k=\omega\sqrt{\varepsilon\mu}$

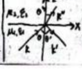
① $\vec E\perp\vec k$, 电场为横波 ② $\vec B\perp\vec k$ 磁场为横波

③ $\vec E\vec B$同相. $\frac{|\vec E|}{|\vec B|}=v=c$.

能量密度 $w=\varepsilon E^2=\frac1\mu B^2$. 能流密度 $\vec S=\vec E\times\vec H=\sqrt{\frac\varepsilon\mu}E^2\hat k=wv$

计算w, S平均时, 取E, B实数部分. $\bar f\bar g=\frac12\mathrm{Re}(f^*g)$

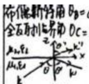
平均值 $\bar w=\frac12\varepsilon E_0^2=\frac{1}{2\mu}B_0^2$ $\bar S=\frac12(\vec E^*\times\vec H)=\frac12\sqrt{\frac\varepsilon\mu}E_0^2\hat k$.



$\begin{cases}k_x=k_x'=k_x''\\ k_y=k_y'=k_y''\\ \omega=\omega'=\omega''\end{cases}$ $\begin{cases}r_s=-\frac{\sin(\theta-\theta'')}{\sin(\theta+\theta'')}\\ t_s=1+r_s\\ r_p=\frac{\tan(\theta-\theta'')}{\tan(\theta+\theta'')}\\ t_p=(1+r_p)\frac{\mu_1}{\mu_2}\end{cases}$

布儒斯特角 $\theta_B=\arctan\frac{n_2}{n_1}$, $r_p=0$. $n=\sqrt{\varepsilon_r}$

全反射临界角 $\theta_C=\arcsin\frac{n_2}{n_1}$ $(n_1>n_2)$



$k_x''=k_x=k\sin\theta$ $k_z''=i\sqrt{k_x^2-k''^2}=ik\sqrt{\sin^2\theta-(\frac{n_2}{n_1})^2}$ $=ik$ (透射深度为 $\frac1\kappa$)

折射波传递因子 $e^{i\vec k''\cdot\vec x}=e^{ikx\sin\theta}\cdot e^{-\kappa z}$, $\bar S_z''=0$

沿z方向衰减, 沿x方向传播. 透射波全部回复

$r_\perp=e^{i\delta_\perp}$ $\tan\frac{\delta_\perp}{2}=-\frac{\sqrt{n_1^2\sin^2\theta-n_2^2}}{n_1\cos\theta}$

$r_\parallel=e^{i\delta_\parallel}$ $\tan\frac{\delta_\parallel}{2}=-\frac{n_1\sqrt{n_1^2\sin^2\theta-n_2^2}}{n_2^2\cos\theta}$

良导体条件: $\frac{\sigma}{\varepsilon\omega}\gg1$. $\frac1\omega\gg\tau$.

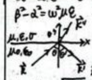
$\nabla\times\vec E=i\omega\mu\vec H$, $\nabla\cdot\vec E=0$, $\nabla\times\vec H=-i\omega\varepsilon\vec E+\sigma\vec E$, $\nabla\cdot\vec H=0$

复电容率 $\varepsilon'=\varepsilon+i\frac\sigma\omega$, $\nabla\times\vec H=-i\omega\varepsilon'\vec E$.

$k^2=\omega^2\mu\varepsilon'$ $\nabla^2\vec E+k^2\vec E=0$. $\vec k=\vec\alpha+i\vec\beta$

$\begin{cases}\vec E_i=\vec E_0e^{i(k_xx+k_zz-\omega t)}\\ \vec E_t=\vec E_0^te^{-\alpha_zz}e^{i(\beta_xx+\beta_zz-\omega t)}\end{cases}$ $k_x^{(t)}=k_x=\beta_x$, $\alpha_x=0$ $k_y^{(t)}=k_y=\beta_y+i\alpha_y=0$

$\beta^2-\alpha^2=\omega^2\mu\varepsilon$ $\vec\alpha\cdot\vec\beta=\frac12\omega\mu\sigma$. $\beta_x=k_x^{(t)}=k\sin\theta$



$\beta^2-\alpha^2=\beta_x^2+\beta_z^2-\alpha_z^2=\omega^2\mu\varepsilon$ $\vec\alpha\cdot\vec\beta=\alpha_z\beta_z=\frac12\omega\mu\sigma$. $\delta=\frac{1}{\alpha_z}$

非垂直入射时, 对良导体有,

$\alpha_z\approx\beta_z\approx\sqrt{\frac{\omega\mu\sigma}{2}}$, $\beta_x\ll\beta_z$ $\delta=\frac{1}{\alpha_z}=\sqrt{\frac{2}{\omega\mu\sigma}}$.

$R=1-2\sqrt{\frac{2\omega\varepsilon_0}{\sigma}}=1-\frac{2\omega}{c}\delta$ (垂直入射) $E+E'=E''$, $H+H'=H''$

谐振腔. 表面处 $E_t=0$, $\frac{\partial E_n}{\partial n}=0$.

$\begin{cases}E_x=A_1\cos k_xx\sin k_yy\sin k_zz\,e^{-i\omega t}, & k_x=\frac{m\pi}{L_1}\\ E_y=A_2\sin k_xx\cos k_yy\sin k_zz\,e^{-i\omega t}, & k_y=\frac{n\pi}{L_2}\\ E_z=A_3\sin k_xx\sin k_yy\cos k_zz\,e^{-i\omega t}, & k_z=\frac{p\pi}{L_3}\end{cases}$

$k_xA_1+k_yA_2+k_zA_3=0$. 两个独立偏振波模.

$k^2=k_x^2+k_y^2+k_z^2=\omega^2\mu\varepsilon$. $\omega_{mnp}=\frac{\pi}{\sqrt{\mu\varepsilon}}\sqrt{(\frac{m}{L_1})^2+(\frac{n}{L_2})^2+(\frac{p}{L_3})^2}$

主模 110: $f_{110}=\frac c2\sqrt{\frac{1}{L_1^2}+\frac{1}{L_2^2}}$ $\lambda_{110}=\frac{2L_1L_2}{\sqrt{L_1^2+L_2^2}}$ $(L_1>L_2>L_3)$

波导表面 $E_t=0$, $\frac{\partial E_n}{\partial n}=0$ 12个边界条件 TE10 TE01 TM11

$\begin{cases}E_x=A_1\cos k_xx\sin k_yy\,e^{i(k_zz-\omega t)}, & k_x=\frac{m\pi}{a}\\ E_y=A_2\sin k_xx\cos k_yy\,e^{i(k_zz-\omega t)}, & k_y=\frac{n\pi}{b}\\ E_z=A_3\sin k_xx\sin k_yy\,e^{i(k_zz-\omega t)}, & k_z=k^2-k_x^2-k_y^2\end{cases}$

$A_3=\frac{A_1k_x+A_2k_y}{ik_z}$ m,n不能同时为0.

TE: $E_z=0$, $H_z\neq0$. TM: $H_z=0$, $E_z\neq0$. (m,n均不为0)

$\lambda>\frac{2}{\sqrt{(\frac ma)^2+(\frac nb)^2}}$, $\omega_{c,mn}=\frac{\pi}{\sqrt{\mu\varepsilon}}\sqrt{(\frac ma)^2+(\frac nb)^2}$, $\omega>\omega_{c,mn}$ $k_z>0$.

电磁波辐射

库仑规范 $\nabla\cdot\vec A=0$ $\nabla\times\vec A=\vec B$, $\nabla(\nabla\cdot\vec A)-\nabla^2\vec A=\mu\vec J$

洛伦兹规范 $\nabla\cdot\vec A+\frac{1}{c^2}\frac{\partial\varphi}{\partial t}=0$ $-\nabla\varphi=\vec E+\frac{\partial\vec A}{\partial t}$

$\begin{cases}\nabla^2\vec A-\frac{1}{c^2}\frac{\partial^2\vec A}{\partial t^2}-\nabla(\nabla\cdot\vec A+\frac{1}{c^2}\frac{\partial\varphi}{\partial t})=-\mu\vec J\\ \nabla^2\varphi+\frac{\partial}{\partial t}\nabla\cdot\vec A=-\frac{\rho}{\varepsilon_0}\end{cases}$ 达朗贝尔方程

洛伦兹规范下

$\begin{cases}\varphi(\vec x,t)=\frac{1}{4\pi\varepsilon_0}\int\frac{\rho(\vec x',t-\frac rc)}{r}dV\\ \vec A(\vec x,t)=\frac{\mu_0}{4\pi}\int\frac{\vec J(\vec x',t-\frac rc)}{r}dV\end{cases}$, $r=|\vec x-\vec x'|$.

泰勒展开 $\vec A(\vec x)=\frac{\mu_0e^{ikR}}{4\pi R}\int_V\vec J(\vec x')[1-ik\hat e_R\cdot\vec x'+\cdots]dV'$

电偶极辐射 $\vec A^{(0)}(\vec x)=\frac{\mu_0e^{ikR}}{4\pi R}\int\vec J(\vec x')dV'=\frac{\mu_0e^{ikR}}{4\pi R}\dot{\vec p}$

$\vec B=\frac{e^{ikR}}{4\pi\varepsilon_0c^3R}\ddot{\vec p}\times\hat e_R$ $\vec E=\frac{ic}{k}\nabla\times\vec B=\frac{e^{ikR}}{4\pi\varepsilon_0c^2R}(\ddot{\vec p}\times\hat e_R)\times\hat e_R$

球面波, 振幅反比于R

$\bar{\vec S}=\frac12\mathrm{Re}(\vec E^*\times\frac{\vec B}{\mu_0})=\frac{c}{2\mu_0}|\vec B|^2\hat e_R$, 辐射角分布 $f(\theta,\varphi)=\frac{\bar{\vec S}\cdot d\vec S}{d\Omega}=\bar S R^2\hat e_R$

辐射功率 $\bar P=\oint\bar{\vec S}\cdot d\vec S$

特别的, $\vec p=p\hat e_z$, $\vec B=\frac{e^{ikR}}{4\pi\varepsilon_0c^3R}\ddot p\sin\theta\,\hat e_\phi$. $\vec E=\frac{e^{ikR}}{4\pi\varepsilon_0c^2R}\ddot p\sin\theta\,\hat e_\theta$ $\ddot p$为振幅值

$\bar{\vec S}=\frac{1}{32\pi^2\varepsilon_0c^3R^2}|\ddot p|^2\sin^2\theta\,\hat e_R$ $f(\theta,\varphi)=\frac{1}{32\pi^2\varepsilon_0c^3}|\ddot p|^2\sin^2\theta$ $(\theta=\frac\pi2$ 辐射最强$)$

$\bar P=\frac{1}{12\pi\varepsilon_0c^3}|\ddot p|^2$. 若 $\vec p=p_0e^{i\omega t}\hat e_z$, 则 $|\ddot p|^2=p_0^2\omega^4$

短天线辐射 $\dot{\vec p}=\int_V\vec J dV'=\int Id\vec l=\int_{-\frac l2}^{\frac l2}I(z)dz=\frac{I_0l}{2}\hat e_z$, $\ddot{\vec p}=\frac{i\omega}{2}I_0l\hat e_z$

$P=\frac{\mu_0I_0^2\omega^2l^2}{48\pi c}=\frac{\pi}{12}\sqrt{\frac{\mu_0}{\varepsilon_0}}I_0^2(\frac l\lambda)^2$ $(l\ll\lambda$时$)$. $P=\frac12R_rI_0^2$.

辐射电阻 $R_r=\frac{\pi}{6}\sqrt{\frac{\mu_0}{\varepsilon_0}}(\frac l\lambda)^2=197(\frac l\lambda)^2$. $P=\oint\bar{\vec S}\cdot d\vec\sigma=\frac{\omega^4p_0^2\mu_0}{32\pi^2c}\int_0^{2\pi}\int_0^\pi\frac{\sin^2\theta}{R^2}R^2\sin\theta d\theta d\phi$

$\nabla\cdot[\vec E_0\sin(\vec k\cdot\vec r)]=\vec k\cdot\vec E_0\cos(\vec k\cdot\vec r)$ $\nabla\times[\vec E_0\sin(\vec k\cdot\vec r)]=\vec k\times\vec E_0\cos(\vec k\cdot\vec r)$

$\varphi=\frac{1}{4\pi\varepsilon_0}(\frac{Q}{\sqrt{R^2+a^2-2Ra\cos\theta}}-\frac{R_0Q/a}{\sqrt{R^2+(\frac{R_0^2}{a})^2-2R\frac{R_0^2}{a}\cos\theta}})$ $\frac{\partial}{\partial x}(\frac1R)=-\frac{x}{R^3}$. $\frac{\partial^2}{\partial x^2}(\frac1R)=\frac{3x^2-R^2}{R^5}$

$\frac{\partial^2}{\partial x\partial y}=\frac{3xy}{R^5}$

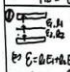
$\begin{cases}\vec B=-\frac{i}{\omega}\nabla\times\vec E=-\frac ik\sqrt{\varepsilon\mu}\nabla\times\vec E\\ \vec E=\frac{i}{\omega\mu\varepsilon}\nabla\times\vec B=\frac{i}{k\sqrt{\mu\varepsilon}}\nabla\times\vec B\end{cases}$ $\begin{cases}\vec H=\sqrt{\frac\varepsilon\mu}\hat k\times\vec E\\ \vec E=-\sqrt{\frac\mu\varepsilon}\hat k\times\vec H\end{cases}$ 底数 k^2 电偶极 $\varphi=\frac{\vec p\cdot\vec r}{4\pi\varepsilon_0r^3}$

$\bar S_z''=\bar{\vec S}''\cdot\hat n=\frac12\mathrm{Re}[\hat n\cdot(\vec E''\times\vec H''^*)]=\frac{1}{2\omega\mu_2}\mathrm{Re}[(\hat n\cdot\vec k'')|\vec E_0''|^2]=0$

$\frac{|\vec H|}{|\vec E|}=\sqrt{\frac{\sigma}{\omega\mu}}\gg1$ 金属电磁波能量主要是磁场能.

电场 $\vec E=\frac{3(\vec p\cdot\vec r)\vec r}{r^5}-\frac{\vec p}{r^3}$

波导 $\begin{cases}H_x=-\frac{i}{\omega\mu k_z}[A_1k_xk_y+A_2(k_y^2+k_z^2)]\sin k_xx\cos k_yy\,e^{i(k_zz-\omega t)}\\ H_y=\frac{i}{\omega\mu k_z}[A_1(k_x^2+k_z^2)+A_2k_xk_y]\cos k_xx\sin k_yy\,e^{i(k_zz-\omega t)}\\ H_z=\frac{1}{\omega\mu}[A_1k_y-A_2k_x]\cos k_xx\cos k_yy\,e^{i(k_zz-\omega t)}\end{cases}$ $\nabla(\vec p\cdot\vec r)=\vec p$ $\nabla\frac1r=-\frac{\vec r}{r^3}$ $\nabla^2\varphi=-\frac{(\vec p\cdot\nabla)\delta(\vec x)}{\varepsilon_0}$.

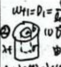


① (1) $\vec D_1=\varepsilon_1\varepsilon_0\hat e_n$, $\vec D_2=\varepsilon_2\varepsilon_0\hat e_n$. $\omega_{f1}=D_1$, $\omega_{f2}=-D_2$. 绝缘介质分界面无自由电荷. $D_2-D_1=\omega_{f3}=0$, $D_1=D_2$ $\begin{cases}\omega_{f1}=\frac{\varepsilon_1\varepsilon_2}{l_1\varepsilon_2+l_2\varepsilon_1}\mathscr E\\ \omega_{f2}=-\frac{\varepsilon_1\varepsilon_2}{l_1\varepsilon_2+l_2\varepsilon_1}\mathscr E\\ \omega_{f3}=0.\end{cases}$

$\mathscr E=l_1E_1+l_2E_2=l_1\frac{D}{\varepsilon_1}+l_2\frac{D}{\varepsilon_2}$

(2) $\mathscr E=l_1E_1+l_2E_2=J_n(\frac{l_1}{\sigma_1}+\frac{l_2}{\sigma_2})$. $J_n=\frac{\sigma_1\sigma_2}{l_1\sigma_2+l_2\sigma_1}\mathscr E$

$\omega_{f1}=D_1=\frac{\varepsilon_1\sigma_2\mathscr E}{l_1\sigma_2+l_2\sigma_1}$ $\omega_{f2}=-D_2=\frac{-\varepsilon_2\sigma_1\mathscr E}{l_1\sigma_2+l_2\sigma_1}$ $\omega_{f3}=D_2-D_1=\frac{\varepsilon_2\sigma_1-\varepsilon_1\sigma_2}{l_1\sigma_2+l_2\sigma_1}\mathscr E$



② (1) $\vec D=\frac{\lambda_f}{2\pi r}\hat e_r$ $\vec E=\frac{\lambda_f}{2\pi\varepsilon r}\hat e_r$ $\vec J_f=\sigma\vec E=\frac{\sigma\lambda_f}{2\pi\varepsilon r}\hat e_r$. $\vec J_D=\frac{\partial\vec D}{\partial t}=\frac{1}{2\pi r}\frac{\partial\lambda_f}{\partial t}\hat e_r$

由 $\nabla\cdot\vec D=\rho_f$. $\nabla\cdot\vec J_f+\frac{\partial\rho_f}{\partial t}=0$ 得 $\frac{\partial\lambda_f}{\partial t}=-\frac\sigma\varepsilon\lambda_f$. 故 $\vec J_f+\frac{\partial\vec D}{\partial t}=0$, $\nabla\times\vec B=$

(2) $\lambda_f(t)=\lambda_f(0)e^{-\frac\sigma\varepsilon t}$ (3) 能量耗散密度 $p=\vec J_f\cdot\vec E=\sigma(\frac{\lambda_f}{2\pi\varepsilon r})^2$

(4) $\int_a^b\sigma(\frac{\lambda_f}{2\pi\varepsilon r})^22\pi rl\,dr=\frac{\sigma\lambda_f^2l}{2\pi\varepsilon^2}\ln\frac ba$ $w=\frac12\vec E\cdot\vec D$. $\frac{\partial w}{\partial t}=-\sigma(\frac{\lambda_f}{2\pi\varepsilon r})^2$.

$-\int_V\frac{\partial w}{\partial t}dV=\frac{\sigma l\lambda_f^2}{2\pi\varepsilon^2}\ln\frac ba$. 电流热效应耗散能量全部由电场能提供.

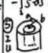
③ (1) $q=\int_0^tIdt=\frac{I_0}{\omega}\sin\omega t$ $\vec E_1=\frac{\sigma}{\varepsilon_0}\hat e_z=\frac{q}{\varepsilon_0\pi a^2}\hat e_z=\frac{I_0}{\pi a^2\varepsilon_0\omega}\sin\omega t\,\hat e_z$

$I=I_0\cos\omega t$ 由 $\oint_L\vec B\cdot d\vec l=\mu_0\varepsilon_0\int\frac{\partial\vec E}{\partial t}\cdot d\vec S$ 得 $\vec B_{in}=\frac{\mu_0I_0r}{2\pi a^2}\cos\omega t\,\hat e_\phi$, $\vec B_{out}=\frac{\mu_0I_0}{2\pi r}\cos\omega t\,\hat e_\phi$

(2) $\vec\alpha_f=\hat e_z\times(\vec H_2-\vec H_1)=-\frac{I_0}{2\pi}(\frac1r-\frac{r}{a^2})\cos\omega t\,\hat e_r$.

④ $\vec E=\frac{I}{\sigma\pi a^2}\hat e_z$. $\vec H=\frac{Ir}{2\pi a^2}\hat e_\phi$ $(r<a)$. $\vec H=\frac{I}{2\pi r}\hat e_\phi$ $(r>a)$. $\vec S=\vec E\times\vec H|_{r=a}=-\frac{I^2}{2\pi^2\sigma a^3}\hat e_r$

$-\oint\vec S\cdot d\vec S=\frac{I^2}{2\pi^2\sigma a^3}2\pi al=\frac{I^2l}{\sigma\pi a^2}$, 而 $P=\sigma E^2\pi a^2l=\frac{I^2l}{\sigma\pi a^2}=I^2R$. 二者相等.



⑤ $H_\phi=\frac{I}{2\pi r}$. $E_r=\frac{\tau}{2\pi\varepsilon r}$ $\vec S=\frac{I\tau}{4\pi^2\varepsilon r^2}\hat e_z$, $U=\int_a^bE_rdr=\frac{\tau}{2\pi\varepsilon}\ln\frac ba$. 代入

$\vec S=\frac{UI}{2\pi\ln\frac ba}\frac{1}{r^2}\hat e_z$ $P=\int_a^bS\cdot2\pi rdr=UI$.

⑥ (1) 由 $\nabla\times\vec E=-\mu_0\frac{\partial\vec H}{\partial t}$ 得 $\vec H=\frac{100k}{10^8\mu_0}\cos(10^8t-kz)\hat e_\phi$

(2) $k=10^8\sqrt{\varepsilon_0\mu_0}=\frac13$ rad/m (3) $\vec\alpha_S=\hat e_n\times\vec H|_{\rho=a}=\hat e_\rho\frac{100}{120\pi\rho}\cos(10^8t-\frac z3)$ A/m

(4) $I_D=\int_S\vec J_D\cdot d\vec S=\int_0^1\vec J_D\cdot\hat e_\rho2\pi\rho dz=0.55\sin(10^8t-\frac z3)$ A

$\vec J_D=-\hat e_\rho\frac{2.85\times10^{-2}}{\rho}\sin(10^8t-\frac z3)$ A/m²

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① $\vec P=K\frac{\vec r}{r^2}$. $\rho_P=-\nabla\cdot\vec P=-\frac{K}{r^2}$. $\sigma_P=-\hat e_r\cdot(\vec P_2-\vec P_1)=\frac KR$ $q_P=\int_V\frac{K}{r^2}dV=-4\pi KR$

$q_\sigma=\frac KR(4\pi R^2)=4\pi KR$ $\rho_f=-\frac{\varepsilon}{\varepsilon-\varepsilon_0}\rho_P=\frac{\varepsilon}{\varepsilon-\varepsilon_0}\frac{K}{r^2}$. $q_f=\int\rho_fdV=\frac{4\pi\varepsilon KR}{\varepsilon-\varepsilon_0}$

$\vec E_1=\frac{\vec P}{\varepsilon-\varepsilon_0}=\frac{K}{\varepsilon-\varepsilon_0}\frac{\vec r}{r^2}$, $(r<R)$. $\vec E_2=\frac{q_f}{4\pi\varepsilon_0}\frac{\vec r}{r^3}=\frac{\varepsilon KR}{\varepsilon_0(\varepsilon-\varepsilon_0)}\frac{\vec r}{r^3}$ $(r>R)$

$\varphi_2=\int_r^\infty\vec E_2\cdot d\vec r=\frac{\varepsilon KR}{\varepsilon_0(\varepsilon-\varepsilon_0)r}$ $(r>R)$ $\varphi_1=\int_r^R\vec E_1\cdot d\vec r+\frac{\varepsilon K}{\varepsilon_0(\varepsilon-\varepsilon_0)}=\frac{K}{\varepsilon-\varepsilon_0}\ln\frac Rr+\frac{\varepsilon K}{\varepsilon_0(\varepsilon-\varepsilon_0)}$ $(r<R)$

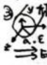
② 假设E沿径向, 左右相同. 设 $\vec E_1=\frac{A}{r^3}\vec r$, $\vec E_2=\frac{A}{r^3}\vec r$ 电场球对称

$\oint\vec D\cdot d\vec S=Q=\varepsilon_1\frac{A}{r^2}2\pi r^2+\varepsilon_2\frac{A}{r^2}2\pi r^2\Rightarrow A=\frac{Q}{2\pi(\varepsilon_1+\varepsilon_2)}$

$\vec E=\begin{cases}\frac{Q}{2\pi(\varepsilon_1+\varepsilon_2)r^3}\vec r, & \text{左半部}\\ \frac{Q}{2\pi(\varepsilon_1+\varepsilon_2)r^3}\vec r, & \text{右半部}\end{cases}$ $\sigma_f=\begin{cases}\frac{\varepsilon_1Q}{2\pi(\varepsilon_1+\varepsilon_2)R_1^2}, & \text{左半部}\\ \frac{\varepsilon_2Q}{2\pi(\varepsilon_1+\varepsilon_2)R_1^2}, & \text{右半部}\end{cases}$ $\sigma_{1p}=-\frac{(\varepsilon_1-\varepsilon_0)Q}{2\pi(\varepsilon_1+\varepsilon_2)R_1^2}$ $\sigma_{2p}=-\frac{(\varepsilon_2-\varepsilon_0)Q}{2\pi(\varepsilon_1+\varepsilon_2)R_1^2}$

设 $\varphi_{\varepsilon_1}=\varphi_0+\frac{P}{r}$, $\varphi_{\varepsilon_2}=\varphi_0+\frac{P}{r}$. 带而 $\varphi_1=\varphi_2$, $\varepsilon_1\frac{\partial\varphi_1}{\partial n}=\varepsilon_2\frac{\partial\varphi_2}{\partial n}$. $\varphi_0=-\frac{P}{R_2}$.

$\oint\varepsilon\vec E\cdot d\vec S=\oint_S\varepsilon_i R_2(-\frac{\partial\varphi}{\partial r})dS=Q=\frac{P}{R_2^2}2\pi R_2^2(\varepsilon_1+\varepsilon_2)$ $\varphi=\frac{Q}{2\pi(\varepsilon_1+\varepsilon_2)}(\frac1r-\frac{1}{R_2})$

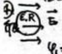


③ 设 $\vec E_0$. 取 $\varphi|_{r=0}=0$. $\varphi=\sum_{n=1}^\infty[r^nB_n\cos n\theta+r^{-n}D_n\cos n\theta]$

$\nabla^2\varphi_1=0$ $(r<a)$. $\nabla^2\varphi_2=0$ $(r>a)$. 由 $r\to\infty$, $\varphi_2\to-E_0r\cos\theta$

得 $\begin{cases}\varphi_1=\sum_{n=1}^\infty B_n^{(1)}r^n\cos n\theta, & 0<r<a\\ \varphi_2=\sum_{n=1}^\infty D_n^{(2)}r^{-n}\cos n\theta-E_0r\cos\theta, & r>a.\end{cases}$ $r=a$时 $\varphi_1=\varphi_2$ $\varepsilon\frac{\partial\varphi_1}{\partial r}|_{r=a}=\varepsilon_0\frac{\partial\varphi_2}{\partial r}|_{r=a}$

得 $\begin{cases}\varphi_1=-\frac{2\varepsilon_0}{\varepsilon+\varepsilon_0}E_0r\cos\theta, & 0<r<a\\ \varphi_2=-E_0r\cos\theta+\frac{\varepsilon-\varepsilon_0}{\varepsilon+\varepsilon_0}\frac{a^2}{r}E_0\cos\theta, & r>a\end{cases}$



④ $\nabla^2\varphi_1=0$ $(r>R)$. $\nabla^2\varphi_2=0$, $(r<R)$ $\varphi_1|_{r\to\infty}=0$. $\varphi_2|_{r=0}$ 有限值.

$\varphi_1|_{r=R}=\varphi_2|_{r=R}$ $\varepsilon_0\frac{\partial\varphi_1}{\partial r}|_{r=R}=\varepsilon\frac{\partial\varphi_2}{\partial r}|_{r=R}$

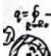
$\varphi_1=\sum_{n=0}^\infty\frac{b_n}{r^{n+1}}P_n(\cos\theta)-E_0r\cos\theta$, $\varphi_2=\sum_{n=0}^\infty a_nr^nP_n(\cos\theta)$

得 $\begin{cases}\varphi_1=-E_0r\cos\theta+\frac{\varepsilon-\varepsilon_0}{2\varepsilon_0+\varepsilon}R^3E_0\frac{1}{r^2}\cos\theta, & r>R\\ \varphi_2=-\frac{3\varepsilon_0}{2\varepsilon_0+\varepsilon}E_0r\cos\theta, & r<R\end{cases}$

⑤ 设 $\varphi(r)=A+\frac Br$. $\nabla^2\varphi_1=0$, $r>R_3$. $\nabla^2\varphi_2=0$, $R_1<r<R_2$.

$\varphi_2|_{r=R_1}=0$. $\varphi_1|_{r\to\infty}=0$. $\varphi_2|_{r=R_2}=\varphi_1|_{r=R_3}$

$-\iint_{r=R_1}\frac{\partial\varphi_2}{\partial r}r^2d\Omega+\iint_{r=R_3}\frac{\partial\varphi_1}{\partial r}r^2d\Omega=\frac{Q}{\varepsilon_0}$ 得 $\begin{cases}\varphi_1=\frac{Q}{4\pi\varepsilon_0r}+\frac{Q_1}{4\pi\varepsilon_0r}\\ \varphi_2=-\frac{Q_1}{4\pi\varepsilon_0R_1}+\frac{Q_1}{4\pi\varepsilon_0r}.\end{cases}$ $Q_1=-\frac{Q}{R_3(\frac{1}{R_1}-\frac{1}{R_2}+\frac{1}{R_3})}$

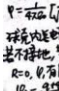


⑥ (1) $\nabla^2\varphi=0$ $(R>R_0)$, $\varphi|_{R\to\infty}\to-E_0R\cos\theta$, $\varphi|_{R=R_0}=\Phi_0$

$\varphi=-E_0R\cos\theta+\sum_n\frac{b_n}{R^{n+1}}P_n(\cos\theta)=-E_0R\cos\theta+\frac{\Phi_0R_0}{R}+\frac{E_0R_0^3}{R^2}\cos\theta$

① Φ_0 ② q (2) $\nabla^2\varphi=0$ $(R>R_0)$ $\varphi|_{R\to\infty}=-E_0R\cos\theta$. $\varphi|_{R=R_0}=\Phi_0$ (未知)

$q=\oint_{R=R_0}-\varepsilon_0\frac{\partial\varphi}{\partial R}ds$, 得 $\varphi=-E_0R\cos\theta+\frac{q}{4\pi\varepsilon_0R}+\frac{E_0R_0^3}{R^2}\cos\theta$.



⑦ 点电荷 $\nabla^2\varphi=\frac{-q\delta(\vec x-a\hat e_z)}{\varepsilon_0}$ $R=0$, φ有限; $R=R_1$, $\varphi=0$

将导体球面感应电荷等效电荷 q'置于 $z=b$中. $b>R_1$.

使 $\varphi=0$, 得 $q'=-\frac{qR_1}{a}$ $b=\frac{R_1^2}{a}$ 因球壳 接地 外表面为0, 感应电荷只能分布在球壳内表面, 这为 $-q$.

$\varphi=\frac{1}{4\pi\varepsilon_0}[\frac{q}{\sqrt{R^2+a^2-2Ra\cos\theta}}+\frac{-qR_1/a}{\sqrt{R^2+(R_1^2/a)^2-2R(R_1^2/a)\cos\theta}}]$

球壳内壁电荷 q_1+q_2, 而 $\vec D=0$. $q+q_2=\oint\vec D\cdot d\vec S=0$, $q_2=-q$.

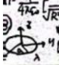
若不接地, 带电荷 φ_0: $\nabla^2\varphi_1=-\frac{q\delta(\vec x-a\hat e_z)}{\varepsilon_0}$ $\nabla^2\varphi_3=0$

$R=0$, φ_1有限, $R\to\infty$, $\varphi_3\to0$. $\varphi_1|_{R=R_1}=\varphi_3|_{R=R_2}=\varphi_0$ $R=R_2$, $q+q_0=\oint\frac{\partial\varphi_3}{\partial R}(-\varepsilon_0)dS$

$\varphi_3=\frac{q+q_0}{4\pi\varepsilon_0R}$ $(R>R_2)$ $\varphi_0=\frac{q+q_0}{4\pi\varepsilon_0R_2}$, $\sigma_2=\frac{q+q_0}{4\pi R_2^2}$ $(R=R_2)$

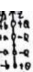
$\varphi_1=\frac{1}{4\pi\varepsilon_0}[\frac{q}{\sqrt{R^2+a^2-2Ra\cos\theta}}+\frac{-qR_1/a}{\sqrt{R^2+(R_1^2/a)^2-2R(R_1^2/a)\cos\theta}}+\frac{q+q_0}{R_2}]$ (已知q)

$\varphi_1=\frac{1}{4\pi\varepsilon_0}[\frac{q}{\sqrt{R^2+a^2-2Ra\cos\theta}}+\frac{-qR_1/a}{\sqrt{R^2+(R_1^2/a)^2-2R(R_1^2/a)\cos\theta}}]+\varphi_0$ (已知 φ_0).



⑧ 电荷分布原点对称, $p=0$ $D_{ij}=\int\rho(3x_ix_j-R_i^2\delta_{ij})dV=\int_0^{2\pi}\lambda(3x_ix_j-r^2\delta_{ij})ad\theta$

$D_{xx}=D_{yy}=\pi a^3\lambda$. $D_{zz}=-2\pi a^3\lambda$. $D_{i\neq j}=0$. $\varphi=\frac{1}{4\pi\varepsilon_0}[\frac{2\pi a\lambda}{R}+\frac{\pi\lambda a^3}{2}\frac{x^2+y^2-2z^2}{R^5}]$

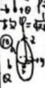


⑨ 中心对称, $p=0$. 净电荷为0. 故 $\varphi^{(0)}=\varphi^{(1)}=0$

$D_{ij}=\int3x_i'x_j'\rho(\vec x')dV'$. $D_{33}=6Q(b-a)=6pl$

$p=Q(b-a)$, $l=b+a$. 其余D均为0.

$\varphi=\frac{1}{4\pi\varepsilon_0}\frac16D_{33}\frac{\partial^2}{\partial z^2}\frac1R=\frac{pl}{4\pi\varepsilon_0}\frac{3z^2-R^2}{R^5}$. $\sum D_{ii}=0$



⑩ $\frac{x^2+y^2}{b^2}+\frac{z^2}{a^2}\le1$. $\rho_0=\frac{3Q}{4\pi ab}$. $\int x_i'x_j'\rho(\vec x')dV'=0$

$\int x^2dV=\int y^2dV=\frac12\int(x^2+y^2)dV=\frac12\int_{-a}^ad\theta\int_0^{b\sqrt{1-\frac{z^2}{a^2}}}2\pi s^3ds$

$=\frac{4\pi ab^4}{15}$

$\int z^2dV=\int_{-a}^az^2dz\int_0^{b\sqrt{1-\frac{z^2}{a^2}}}2\pi sds=\frac{4\pi a^3b^2}{15}$ $+\frac{\vec p\sin\theta\hat e_\theta}{r^3}$

故 $D_{33}=\frac25(a^2-b^2)Q$, $D_{11}=D_{22}=-\frac15(a^2-b^2)Q$

$\varphi=\frac{1}{4\pi\varepsilon_0}(\frac1R+\frac{a^2-b^2}{10}\frac{3\cos^2\theta-1}{R^3})$.

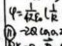
⑪ $-2Q\,(0,0,2a)$, $2Q\,(0,0,a)$, $3Q\,(0,0,-a)$, $-2Q\,(0,0,-2a)$

$Q_{总}=Q$. $\vec p=Qa\hat e_z$. $\varphi(\vec x)=\frac{1}{4\pi\varepsilon_0}(\frac QR-\vec p\cdot\nabla\frac1R)=\frac{1}{4\pi\varepsilon_0}(\frac QR+\frac{Qaz}{R^3})$

即 $\varphi(\vec x)=\frac{Q}{4\pi\varepsilon_0R}(1+\frac{a\cos\theta}{R})$ $\vec E=-\nabla\varphi=\frac{Q\vec R}{4\pi\varepsilon_0R^3}+\frac{Qa}{4\pi\varepsilon_0}(\frac{2\cos\theta\hat R}{r^3}$

①水→空气 $\theta=60°$ 折射波相速度和透入深度.

$\theta_c=\arcsin\frac{1}{1.33}=49°45'<60°$. 空气 $\lambda_0=6.28\times10^{-5}$ cm.



$k''=\frac{k}{1.33}=0.752k$. $k_x''=k_x=0.866k$.

$k_z=ik\sqrt{\sin^2\theta-(\frac{n_2}{n_1})^2}=ik\,0.429$

$\vec E''=\vec E_0''e^{-0.429kz}e^{i(0.866kx-\omega t)}$

$\delta=\frac1\kappa=\frac{6.28\times10^{-5}}{0.429\cdot2\pi\cdot1.33}=1.7\times10^{-5}$ cm $v=\frac{\omega}{k_x''}=0.868c$.

② $\vec E_i=E_{i0}\cos(kz-\omega t)\hat e_x$ 垂直入射理想导体 $z=0$. 求 $\vec E$, $\vec H$, $\vec\alpha$.

$\vec E_i=E_{i0}e^{i(kz-\omega t)}\hat e_x$. $\vec H_i=\sqrt{\frac\varepsilon\mu}E_{i0}e^{i(kz-\omega t)}\hat e_y$. $E_{i0}+E_{r0}=0$

$\vec E_r=E_{r0}e^{i(-kz-\omega t)}\hat e_x$, $\vec H_r=-\sqrt{\frac\varepsilon\mu}E_{r0}e^{i(-kz-\omega t)}\hat e_y$ $E_{r0}=-E_{i0}$

$\vec E_实=2E_{i0}\sin kz\sin\omega t\,\hat e_x$ (实部), $\vec H_实=2\sqrt{\frac\varepsilon\mu}E_{i0}\cos kz\cos\omega t\,\hat e_y$

$\vec\alpha=H_y(\omega t)\hat e_x=2\sqrt{\frac\varepsilon\mu}E_{i0}\cos\omega t\,\hat e_x$ $\to$ n为$-z$方向

③ $\vec k''=(\beta+i\alpha)\hat e_z$, $\vec E=\vec E_0e^{-\alpha z}e^{i(\beta z-\omega t)}$, $\alpha=\beta=\sqrt{\frac{\omega\mu\sigma}{2}}$

$\vec H=\frac{\vec B}{\mu}=\frac{1}{\omega\mu}\vec k''\times\vec E=\sqrt{\frac{\sigma}{\omega\mu}}e^{i\frac\pi4}\hat e_z\times\vec E$

$\bar{\vec S}=\frac12\mathrm{Re}(\vec E^*\times\vec H)=\frac12\sqrt{\frac{\sigma}{2\omega\mu}}E_0^2e^{-2\alpha z}\hat e_z$. $\bar S|_{z=0}=\frac12\sqrt{\frac{\sigma}{2\omega\mu}}E_0^2$

$\bar p=\frac12\mathrm{Re}(\vec J_f^*\cdot\vec E)=\frac\sigma2E_0^2e^{-2\alpha z}$. $\int_0^\infty\bar pdz=\frac12\sqrt{\frac{\sigma}{2\omega\mu}}E_0^2$ //

④ $E_z=E_0\sin\frac{\pi x}{L}\sin\frac{\pi y}{L}\cos\omega t$

$H_x=-\sqrt{\frac{\varepsilon_0}{2\mu_0}}E_0\sin\frac{\pi x}{L}\cos\frac{\pi y}{L}\sin\omega t$ $H_y=\sqrt{\frac{\varepsilon_0}{2\mu_0}}E_0\cos\frac{\pi x}{L}\sin\frac{\pi y}{L}\sin\omega t$

$W=\frac12(\varepsilon_0E^2+\mu_0H^2)$. $\overline{E_z^2}=\frac12E_z^2$

$\langle E_z^2\rangle=\frac{E_0^2}{2}\frac{1}{L^2}\int_0^L\sin^2\frac{\pi x}{L}dx\int_0^L\sin^2\frac{\pi y}{L}dy=\frac{E_0^2}{8}$. $\langle H_x^2\rangle=\langle H_y^2\rangle=\frac{\varepsilon_0E_0^2}{16\mu_0}$

$\langle W\rangle=L^3\frac{E_0^2}{2}(\frac{\varepsilon_0}{4}+\frac{\mu_0}{2}(\cdots))=L^3\frac{\varepsilon_0E_0^2}{8}$

⑤ $\nabla\cdot(\vec E\times\vec H^*)=\vec H^*\cdot(\nabla\times\vec E)-\vec E\cdot(\nabla\times\vec H^*)=i\omega(\mu|\vec H|^2-\varepsilon|\vec E|^2)$

$\int_V\nabla\cdot(\vec E\times\vec H^*)dV=\oint_S(\vec E\times\vec H^*)d\vec S$ 由于边界电场切向为0. 磁场切向为0, 故 $\phi=0$

从而有 $\frac{1}{\mu_0}\int|\vec B|^2dV=\varepsilon_0\int|\vec E|^2dV$.

⑥ 4×10^9 Hz TE10且不系 TE20.

$f_{c,mn}=\frac c2\sqrt{(\frac ma)^2+(\frac nb)^2}$. $f_{c,10}=\frac{c}{2a}$ $f_{c,20}=\frac ca$

$f_{c,10}<f<f_{c,20}\Rightarrow3.75$ cm $<f<7.5$ cm We, Wm

⑦ $a\times b$, TE10. 沿z方向平均功率P. 单位长度场平均能量

$E_x=E_z=H_y=0$. $H_z=H_0\cos\frac{\pi x}{a}e^{i(k_zz-\omega t)}$

$H_x=-\frac{ik_za}{\pi}H_0\sin\frac{\pi x}{a}e^{i(k_zz-\omega t)}$ $E_y=\frac{i\omega\mu a}{\pi}H_0\sin\frac{\pi x}{a}e^{i(k_zz-\omega t)}$

$P=\frac12\int_0^a\int_0^b\mathrm{Re}(\vec E\times\vec H^*)\cdot\hat e_zdxdy=\frac12\int_0^a\int_0^b\mathrm{Re}(-E_yH_x^*)dxdy$

$=\frac{\omega k_za^3b}{4\pi^2}\mu|H_0|^2$

$W_e=\frac\varepsilon4|E|^2$ $W_e=\int_0^a\int_0^b\int_0^1w_edxdydz=\frac\varepsilon4(\frac{\omega\mu a}{\pi})^2|H_0|^2\frac{ab}{2}$

$W_m=\int_0^a\int_0^b\int_0^1w_mdxdydz=\frac\mu4\int_0^a\int_0^b\int_0^1(|H_x|^2+|H_z|^2)dxdydz$

$=\frac{\mu a^3b}{8\pi^2}|H_0|^2k^2$

⑧ $f=30\times10^9$ Hz. 0.7cm×0.4cm 波导.

$\lambda=1$ cm $\lambda_{c,m}=\frac{2\pi c}{\omega_{c,mn}}=\frac{2ab}{\sqrt{m^2b^2+n^2a^2}}$.

$\lambda_{c,10}=1.4$ cm, $\lambda_{c,01}=0.8$ cm. $\lambda_{c,11}=0.69$ cm

故仅能传播 TE10 模.

⑨ $\vec H$满足条件. $\nabla\cdot\vec E(\vec x)=0$, $\nabla\times\vec E(\vec x)=i\omega\mu\vec H(\vec x)$

$\nabla\cdot\vec H(\vec x)=0$, $\nabla\times\vec H(\vec x)=-i\omega\varepsilon\vec E$

$\Rightarrow\nabla^2\vec H+k^2\vec H=0$, $k=\frac\omega v$. $\nabla\cdot\vec H=0$

边界 $\hat e_n\times\vec E|_S=0$, $\hat e_n\times(\nabla\times\vec H)|_S=0$

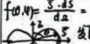
$x=0$, a处, $H_x=0$, $\frac{\partial H_y}{\partial x}-\frac{\partial H_x}{\partial y}=0$.

$y=0$, b处, $H_y=0$, $\frac{\partial H_x}{\partial y}-\frac{\partial H_y}{\partial x}=0$.

$\langle\vec S\rangle=\bar{\vec S}=\frac12\mathrm{Re}(\vec E^*\times\frac{\vec B}{\mu_0})=\frac{1}{2\mu_0}\mathrm{Re}[c(\vec B\times\hat e_R)^*\times\vec B]$

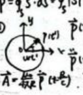
$=\frac{c}{2\mu_0}|\vec B|^2\hat e_R=\frac{|\ddot{\vec p}|^2}{32\pi^2\varepsilon_0c^3R^2}\sin^2\theta\,\hat e_R$

$f(\theta,\varphi)=\frac{\bar{\vec S}\cdot d\vec S}{d\Omega}=\frac{\bar{\vec S}\cdot R^2d\Omega\hat e_R}{d\Omega}=\frac{|\ddot p|^2}{32\pi^2\varepsilon_0c^3}\sin^2\theta$



发射天线水平放置最有效

$\bar P=\oint_S\bar{\vec S}\cdot d\vec S=\oint_S|\bar{\vec S}|R^2\sin\theta d\theta d\varphi=\frac{|\ddot p|^2}{12\pi\varepsilon_0c^3}$ 与R无关



① $\vec p(t')=p_0\cos\omega t'\hat e_x+\sin\omega t'\hat e_y$

$=p_0(\hat e_x+i\hat e_y)e^{-i\omega t'}$

$\ddot{\vec p}(t')=-p_0\omega^2(\hat e_x+i\hat e_y)e^{-i\omega t'}$.

$\vec A=\frac{\mu_0}{4\pi R}\dot{\vec p}(t-\frac Rc)$. $\vec B=\nabla\times\vec A=\frac{1}{4\pi\varepsilon_0c^3R}\ddot{\vec p}\times\hat e_R$

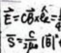
$=\frac{p_0\omega^2}{4\pi\varepsilon_0c^3R}e^{i(kR-\omega t+\varphi)}(-i\hat e_\theta+\cos\theta\hat e_\varphi)$

$\vec E=c\vec B\times\hat e_R=\frac{p_0\omega^2}{4\pi\varepsilon_0c^2R}e^{i(kR-\omega t+\varphi)}(\cos\theta\hat e_\theta+i\hat e_\varphi)$

$\bar{\vec S}=\frac{c}{2\mu_0}|\vec B|^2\hat e_R=\frac{\mu_0p_0^2\omega^4}{32\pi^2cR^2}(1+\cos^2\theta)\hat e_R$

$\frac{dP}{d\Omega}=\langle\vec S\cdot\hat e_R\rangle R^2=\frac{\mu_0p_0^2\omega^4}{32\pi^2c}(1+\cos^2\theta)$

$P=\int\frac{dP}{d\Omega}d\Omega=\frac{p_0^2\omega^4}{6\pi\varepsilon_0c^3}$. $\ddot p=q\dot v=qa$



② Q,ω $\vec p=Qa(\hat e_x+i\hat e_y)e^{-i\omega t}$. $p_0=Qa$

$\vec B=\frac{\mu_0}{4\pi cR}\ddot{\vec p}\times\hat e_R=\frac{Qa\omega^2\mu_0e^{-i\omega t}}{4\pi cR}[(\cos\theta(\cos\phi+i\sin\phi)\hat e_\varphi+(\sin\phi-i\cos\phi)\hat e_\theta]$

$\vec E=c\vec B\times\hat e_R=-\frac{Qa\omega^2\mu_0e^{-i\omega t}}{4\pi R}[-\cos\theta(\cos\phi+i\sin\phi)\hat e_\theta+(\sin\phi+i\cos\phi)\hat e_\phi]$

$\bar{\vec S}=\frac12\mathrm{Re}(\vec E^*\times\vec H)=\frac{Q^2a^2\omega^4\mu_0}{32\pi^2cR^2}(1+\cos^2\theta)\hat e_R$.

$\bar P=\frac{Q^2a^2\omega^4\mu_0}{6\pi c}$

③ 球对称电荷分布, ω径向.

$\vec r'=\vec r_0'e^{-i\omega t}$ $\vec v'=-i\omega\vec r'$. $\vec J(\vec x',t')=\rho(\vec x')\vec v'=-i\omega\rho(\vec x')\vec r'$

$\vec A(\vec x,t)=\frac{\mu_0}{4\pi}\int\frac{\vec J(\vec x',t)}{r}dV'=0$ (E,B均0)

④ R,q,ω: $\lambda=\frac{q}{2\pi R}$. $I=\lambda v=\frac{\omega q}{2\pi}$ 是稳恒的, 无辐射.